State Methodology (11/3)

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Changed on 3 November: 3.4, 4.2, 4.3, 6, and the open items. Changed on 4 November: 6 (home and 2026 price checks).

This appendix explains how I adapt the Budget Lab's national ECEC model to local markets. I use the national methodology's definitions and notation. The baseline is 2019. Each run covers children ages 0–4 in one of the 50 states, DC, or New York City (NYC). NYC is the five boroughs; it is also included in New York State, so I exclude its separate run when adding up national totals.

I keep the nationally estimated forests, utility function, β, ρ, production technology, and care-worker labor-supply elasticities. I change local care-hour targets, wages, staffing requirements, non-labor costs, and parents' employment paths. Each geography is a separate market, with no families or workers moving across its boundary.

1 Income ranks and national calibration

1. **Put incomes on a comparable purchasing-power basis.** I divide 2019 household income by the 2019 BEA all-items regional price parity (RPP), expressed relative to the US average of 100:

$$\text{income used for ranking} = \frac{\text{dollar income}}{\text{all-items RPP}/100}.$$

   An income of \$55,000 in a place with RPP 110 becomes \$50,000 for ranking, because the same dollar income buys less in a more expensive place. The rank enters only the prediction models; the household budget uses dollar income.

2. **Assign the available geographic price index.** ACS households use their state's 2019 BEA all-items RPP. NSECE identifies only four Census regions, so each NSECE household uses its region's average state RPP, weighted by 2019 ACS children under five. NYC uses the New York–Newark–Jersey City metropolitan RPP, 114.9, as a proxy for city prices; the metro area includes parts of New Jersey and Pennsylvania.

3. **Calculate national income ranks within each dataset.** I calculate the 1st through 99th percentile cutoffs of adjusted household income among children under five with positive household income, using child weights. I pool all regions for these cutoffs and calculate them separately in NSECE and ACS. A rank therefore compares a family with families nationally, not only in its own state. The 2019 ranks stay fixed for the trained forests.

4. **Calibrate the utility parameters nationally.** After this income adjustment, I estimate one β and one ρ, shared across geographies and one-child and two-or-more-child parent units. The 26 October 2026 calibration gives $\beta = 0.0003405$ and $\rho = 0.0262383$.

Aggregate elasticity	Target	Achieved
Employment with respect to paid care cost	−0.15	−0.1500
Work hours with respect to after-tax income	−0.05	−0.0500
Paid care hours with respect to paid care cost	—	−1.5137
Employment with respect to caregiver wages	—	1.5636

The experiments raise paid care costs by 1% and add 1% of expected after-tax income as a lump sum, with parent-unit weights. The first two rows are calibration targets; the last two are implied by the model.

2 Center price tiers

1. **Define 36 cutoffs.** I use four Census regions, three community density groups, and three child age bands: 0–1, 2–3, and 4–5. States in the same region use the same cutoff for a given density and age. The same hourly price means a different position in the local market across regions, densities, and ages, so the cutoffs vary. Price is a proxy for quality.

2. **Measure each center's price and enrollment.** For 2019 NSECE center-based providers, the hourly price is the standard weekly posted rate divided by 40, averaged over the single years of age in each band. Medians use positive prices only, weighted by survey weight times full-time-equivalent enrollment in the band, with part-time enrollment counting one-half.

3. **Combine regional and density differences.** Let $\tilde{p}_{r,d,a}$ be the weighted median posted price among centers in region r, density group d, and age band a, with $r = \text{US}$ for all regions pooled. The cutoff is the regional high-density median times the national ratio of density d to high density:

$$\tau_{r,d,a} = \tilde{p}_{r,\text{high},a} \frac{\tilde{p}_{\text{US},d,a}}{\tilde{p}_{\text{US},\text{high},a}}.$$

   I use the national density ratio because the regional rural cells contain only 5–35 surveyed centers per age band, and moderate-density cells contain 28–134 (unweighted counts of centers).

4. **Assign each care arrangement to a tier.** A center uses its own region and density group. A family uses its household's region and density group and its child's age band. I compare posted hourly prices for providers and weekly out-of-pocket payment divided by weekly care hours for households. A price at or above $\tau_{r,d,a}$ is high-price. Household payments are net of third-party support, so their tier can differ from the provider's posted-price tier.

NSECE community density measures the urban share of the nearby population, using tracts intersecting a two-mile circle around the anchor tract's population centroid and inverse-distance weights. High density means at least 85% urban, moderate means 30% to under 85%, and rural means under 30%.

Region	Child age	High density	Moderate	Rural
Northeast	0–1	6.73	6.40	5.09
	2–3	5.75	5.30	3.85
	4–5	5.33	4.52	3.58
Midwest	0–1	5.88	5.60	4.44
	2–3	4.75	4.38	3.18
	4–5	4.38	3.71	2.94
South	0–1	4.49	4.27	3.39
	2–3	3.94	3.63	2.64
	4–5	3.73	3.16	2.51
West	0–1	6.36	6.05	4.81
	2–3	5.29	4.88	3.55
	4–5	4.89	4.15	3.29

Cutoffs are in 2019 dollars per hour, rounded to cents.

3 Local care-hour targets

3.1 Children in regular care

1. **Estimate the state participation rate.** I pool the 2016–2019 NSCH observations for children ages 0–4 with a valid yes/no answer on at least ten weekly hours of non-parental care. The rate is the sum of child survey weights for yes answers divided by the sum for all valid answers. Pooling four pre-pandemic years reduces sampling noise in state estimates. NYC uses New York's rate because NSCH does not identify the city.

2. **Convert the rate to children using the model's care definition.** I multiply the rate π_g by the Census 2019 population ages 0–4, Pop_g (county estimates summed), and by a national factor that converts the NSCH ten-hour definition to NSECE's regular-care definition, which starts at five hours. The factor is the 2019 NSECE child-weighted count of children ages 0–4 in regular non-parental care, excluding school programs, divided by the count with at least ten weekly hours of that care:

$$K_g = \text{Pop}_g \, \pi_g \, (1.098054),$$

where K_g is the number of children in regular care. For the 50 states and DC, the result is 11.6 million children in regular care, 59.2% of 19.6 million children ages 0–4. These are population estimates, not unweighted survey counts.

3.2 Children in unpaid centers

1. **Pre-K.** I take the larger of public-school pre-K enrollment from the Common Core of Data and all-settings state pre-K enrollment from Table 2 of the 2019 NIEER yearbook. NYC uses its own all-settings city count. The larger count covers public-school programs outside the state program and state programs outside public schools, without double counting their overlap.

2. **Head Start and Early Head Start.** I convert funded slots to point-in-time enrolled children using national 2018–19 Office of Head Start enrollment-to-slot ratios: about 0.957

for Head Start and 0.923 for Early Head Start. For Early Head Start, I first remove the 3.592% share serving pregnant women. I remove home-visiting and family-child-care options using national program shares of about 1.94% for Head Start and 38.0% for Early Head Start.

3. **Remove children outside the modeled ages.** National 2019 NHES child-weighted shares, with age measured on December 31, remove 24.40% of pre-K children and 10.74% of Head Start children as age five or older. Early Head Start needs no age-five adjustment. NYC's pre-K calendar-year eligibility already keeps its count within ages 0–4, and NYC Head Start serves no age-five children, so both NYC shares are zero. For New York State, I apply the national pre-K and Head Start shares only to the enrollment outside NYC.

4. **Remove pre-K/Head Start overlap.** I calculate one pooled overlap share o: the Head-Start-slot-weighted mean of the overlap shares in the 10 state/DC geographies that use the all-settings pre-K count and have NIEER unduplicated counts (AL, CT, DC, GA, IA, MA, MD, NC, NJ, and NY). I apply this pooled share, $o = 27.52\%$, to adjusted Head Start in every geography:

$$K_{g,\mathrm{UC}} = \text{adjusted pre-K}_g + (1 - o)\text{adjusted Head Start}_g + \text{adjusted Early Head Start}_g,$$

where $K_{g,\mathrm{UC}}$ is the number of children in unpaid centers. State counts alone cannot identify a child in both programs, so the pooled share estimates the overlap. $K_{g,\mathrm{UC}}$ can still overstate the model's unpaid-center category when programs charge fees or children also attend paid care.

3.3 Allocate children and convert to annual hours

1. **Predict the local care mix.** I apply the national model to the geography's ACS families. For each care sector s, I sum each child's probability of that sector times the child weight to obtain $B_{g,s}$. I exclude parental care and combine the two paid-center tiers into one sector for this step. These are predicted population counts of children.

2. **Allocate the remaining children.** After setting unpaid-center children to $K_{g,\mathrm{UC}}$, I distribute $K_g - K_{g,\mathrm{UC}}$ across paid center, paid home, unpaid home, other paid, and other unpaid care in proportion to their B values:

$$K_{g,s} = (K_g - K_{g,\mathrm{UC}})\frac{B_{g,s}}{\sum_{s'} B_{g,s'}},$$

where s' runs over these five sectors. The participation rate and program counts set the level of care; the national model's predictions for local families set the split across sectors.

3. **Calculate six annual-hour targets.** For each sector, including unpaid centers, I use the predicted part-time share $q_{g,s} = B_{g,s,\mathrm{PT}}/(B_{g,s,\mathrm{PT}} + B_{g,s,\mathrm{FT}})$, where $B_{g,s,\mathrm{PT}}$ and $B_{g,s,\mathrm{FT}}$ split $B_{g,s}$ by care hours. At 10 weekly hours for part-time care or 40 for full-time care, for 50 weeks, the target in annual child-hours is

$$\bar{Q}_g^s = K_{g,s}\,[500 q_{g,s} + 2{,}000(1 - q_{g,s})].$$

The model's hour grid makes the targets comparable with simulated care hours.

3.4 Fit local baseline preferences

1. **Express the national preferences at local 2019 prices.** The national model backs out each family's intercepts from its baseline probabilities P^0_{ij}, with $\alpha_{i1} = 0$: $\alpha_{ij} = (\ln P^0_{ij} - \ln P^0_{i1}) - (\tilde{V}_{ij} - \tilde{V}_{i1})$, where $\tilde{V}_{ij} = V_{ij} - \alpha_{ij}$ is the financial part of utility. I apply the same formula with care costs at the geography's 2019 supply prices, $\tilde{V}^g_{ij}$:

$$\alpha^g_{ij} = (\ln P^0_{ij} - \ln P^0_{i1}) - (\tilde{V}^g_{ij} - \tilde{V}^g_{i1}).$$

   At local prices, these intercepts give back P^0_{ij}.

2. **Shift the national intercepts to hit local targets.** Within the geography, I add one utility shift per care sector, a_s, and one per caregiver group, b_k (sex by under 25, 25–34, 35–44, and 45+). An option's intercept rises by the shift of each child's sector, plus the caregiver's group shift if the caregiver works:

$$\alpha'_{ij} = \alpha^g_{ij} + \sum_c a_{s_c(j)} + b_{k(i)} E_j,$$

   where $s_c(j)$ is child c's sector under option j, $k(i)$ is the caregiver's group, and $E_j = 1$ if the caregiver works. Parental care has no shift, and the two paid-center tiers share one. I then subtract option 1's value from every option, so α'_{i1} stays 0. The employment shift has the same form as the national Δ_t. β and ρ stay national.

3. **Solve the shifts by target-to-model ratios.** Starting from zero, I add the log of a target-to-model ratio to each shift:

$$a_s \leftarrow a_s + \ln \frac{\text{target hours}_s}{\text{model hours}_s}, \qquad b_k \leftarrow b_k + \ln \frac{\text{target employment odds}_k}{\text{model employment odds}_k}.$$

   Adding the log of a ratio to an option's intercept multiplies its probability by that ratio before renormalizing. Renormalizing moves the other shares, so I repeat until every log ratio is below 10^{-8}.

   Target hours are the six $\bar{Q}^s_g$ above. Model hours add up each child's expected annual hours in the sector over the geography's ACS families, with child weights. A group's model employment rate is its caregivers' mean probability of working, with person weights; its target is the same rate at P^0, the employment forest's prediction. The build stops if a care sector has no 2019 probability in the geography, because no shift can then reach a positive target.

4. **Fit employment and care together.** Families choose care and work jointly in P^0, so a care shift alone moves employment away from the forest rate. The joint fit holds 2019 employment at the forest rate while care hours move to their targets. Over the 50 states and DC, fits converge in 32 to 152 iterations. Median care shifts are 0.37 to 0.44 by sector, and median employment shifts are -0.22 to -0.09 by group.

5. **Keep the national differences across families.** Every family in the geography gets the same shifts, because local data give totals only. A shift changes the odds between two sectors by the same factor for every family, so differences by income, age, and family composition come from the national model. Options that take the same shift keep their national odds, for example part-time and full-time care in one sector, or low- and high-price centers.

6. **Carry the shifts into later years.** The shifts become part of α', so every simulation year and every policy scenario uses them. In each later year, the national employment shift Δ_t then moves employment only by its projected change since 2019 (section 5).

7. **Use the fitted baseline for policy comparisons.** I hold these preferences fixed when policy changes. Because I fit hours, child counts and part-time shares need not equal those used to build the targets. The model determines the split between low- and high-price centers.

4 Local supply inputs

4.1 Care-worker wages

1. **Fix the occupation mix by worker education.** I use national 2015–2019 ACS workers living in households, with positive earnings and usual work hours. Childcare workers exclude private-household employment; teachers and administrators must work in child day-care services. Preschool special-education teachers count as preschool teachers. Person-weighted shares of childcare workers, preschool teachers, and administrators are 72.6%, 24.2%, and 3.2% without a BA, and 42.6%, 45.8%, and 11.6% with a BA.

2. **Calculate a local wage index.** I take the weighted mean of BLS OEWS median hourly wages for childcare workers (39-9011), preschool teachers (25-2011), and preschool/daycare administrators (11-9031), using the fixed ACS weights. $R_{g,e}$ is this local mean divided by the national mean, for geography g and education group e. The local 2019 wage is the national 2019 model wage times $R_{g,e}$:

$$w_{g,e,2019} = w_{\mathrm{US},e,2019} R_{g,e}, \qquad w_{\mathrm{US},e,2019} = \begin{cases} 12.06 & \text{without BA,} \\ 19.70 & \text{with BA.} \end{cases}$$

   Fixed occupation weights keep differences in occupation mix out of the index. NYC uses the New York metro area, including its New Jersey and Pennsylvania parts.

3. **Project wages.** I multiply the local 2019 wage by national nominal hourly-wage growth and by the change since 2019 in $A_{g,e,y}$, the annual relative wage. A uses 2019–2025 BLS data and the same weights as R, but drops an occupation from the whole path, locally and nationally, if BLS suppresses it in any year, and renormalizes the other weights. $A_{g,e,2019}$ can therefore differ from $R_{g,e}$; only its change enters the wage. After 2025 I hold the relative wage fixed and continue national wage growth.

4.2 Staff hours per child-hour

1. **Measure observed staffing by region and density.** I use the 2019 NSECE provider surveys, with the high, moderate, and rural density groups of the price tiers. For centers, the classroom module gives each classroom's teachers and assistants, enrolled children, and the age of its youngest child. I compare each classroom with national staffing at that age, so a value of 1 means national staffing for the same ages:

$$\hat{\sigma}_{r,d,\mathrm{center}} = \frac{\sum_\ell \omega_\ell \, \mathrm{staff}_\ell}{\sum_\ell \omega_\ell \, \mathrm{enrolled}_\ell \, n_{a(\ell)}},$$

   where ℓ runs over classrooms in region r and density group d, ω_ℓ is the classroom weight, and n_a is national staff per enrolled child at age a: 0.332, 0.260, 0.191, 0.160, and 0.144 at ages

0 through 4. For homes, $\hat{\sigma}_{r,d,\text{home}}$ is the respondent plus helpers per enrolled child in listed homes, with provider weights. Every one of the 24 cells has a relative standard error of 0.30 or less.

	Centers (1 = national)			Homes (staff per child)		
Region	High	Moderate	Rural	High	Moderate	Rural
Northeast	1.061	0.943	1.131	0.205	0.198	0.167
Midwest	1.059	0.899	1.312	0.188	0.191	0.153
South	0.958	1.070	0.962	0.221	0.240	0.190
West	0.949	0.912	0.949	0.201	0.168	0.164

2. **Apply each geography's density mix.** A geography takes its region's cells at its 2010 Census mix of children under five by density, $m_{g,d}$; each tract's urban share stands in for NSECE's two-mile circle:

$$\sigma_{g,s} = \sum_d m_{g,d}\,\hat{\sigma}_{r(g),d,s}, \qquad I_{g,s} = \sigma_{g,s}/\bar{\sigma}_s,$$

where $\bar{\sigma}_s$ is the mean over the 50 states and DC weighted by Census 2019 population ages 0–4. NYC uses the Northeast cells at its own mix and is excluded from the mean. I multiply national staff hours per child-hour by the index, for both education groups. The center index applies to low-price, high-price, and unpaid centers. Center indices run from 0.937 to 1.134.

3. **Blend the home index by the licensed share.** The home cells cover listed (licensed) homes only. Licensed homes hold $\lambda = 25.7\%$ of children ages 0–5 in paid home-based care (2019 NSECE home-based provider survey, weighted). Unlicensed paid home care keeps the national staffing level:

$$I_{g,\text{home}} = \lambda\,\frac{\sigma_{g,\text{home}}}{\bar{\sigma}_{\text{home}}} + (1-\lambda).$$

Home indices run from 0.965 to 1.024.

4.3 Non-labor costs

1. **Start with each national residual.** In the national model, the per-unit residual revenue $\bar{\delta}$ of a sector is its 2019 NSECE price minus its labor cost: \$0.510055 per child-hour for low-price centers, \$4.614567 for high-price centers, and \$2.827343 for paid homes. It includes non-wage costs and provider margins. Unpaid centers have no residual.

2. **Set a local 2019 price.** I compute the local residual the same way, from a local price and the local labor cost. For centers and for paid homes, I take NDCP 2019 county median full-time weekly prices for infants, toddlers, and preschoolers. I weight counties by Census 2019 population ages 0–4 and age groups by the geography's model children in the setting. NDCP gives the geographic pattern. I keep the NSECE national level, because β and ρ are calibrated at NSECE prices:

$$p_g = \frac{\text{NDCP weekly price}_g}{40} \cdot \frac{\text{NSECE national price}}{\text{NDCP national price}},$$

with both national prices child-weighted over the 50 states and DC. The level factor is 1.080 for centers and 1.366 for homes.

3. **Fill missing prices.** Colorado, DC, and Nevada have no 2019 NDCP price, and Mississippi has no 2019 home price. Each takes the nearest NDCP year in 2016–2022, scaled by the national change in each age group's price. Indiana and New Mexico have no NDCP price in 2016–2022. They take the NSECE 2019 price by region and density group at the state's density mix, as for staffing. A price cell with a relative standard error above 0.30, or under one design degree of freedom, takes the national density price times the region-to-national price ratio.

4. **Subtract the local labor cost.** The labor cost is local wages times local staff hours per child-hour, summed over the two education groups. For centers, one factor μ_g scales both national tier residuals, so the local tier prices average to the local center price at the P^0 tier mix. The home residual is the remainder:

$$\mu_g = \frac{p_{g,\text{center}} - \text{labor cost}_{g,\text{center}}}{\bar{\delta}_{\text{US,center}}}, \qquad \bar{\delta}_{g,\text{home}} = p_{g,\text{home}} - \text{labor cost}_{g,\text{home}},$$

   where the center labor cost and national center residual are averages over the two tiers at the P^0 tier mix. μ_g runs from 0.091 to 2.523, median 0.990. The home residual is 0.020 to 2.804 times the national one, median 0.920. The smallest, Mississippi's, is small because observed staffing gives its homes a high labor cost; it stays above zero. The build stops if a residual is zero or less.

5. **Apply annual inflation.** CPI increases the nominal residual over time, consistently with the national model: $\bar{\delta}_{g,y} = \bar{\delta}_{g,2019}\,\text{CPI}_y/\text{CPI}_{2019}$.

4.4 Size the care-worker labor market

I use the national wage equation with local inputs. Required staff hours $L_{g,e,y}$ are the model's care hours in each sector times the local staff hours per child-hour, summed over sectors. With G_y national nominal hourly-wage growth since 2019, the worker wage is

$$w_{g,e,y} = w_{g,e,2019}\,G_y\,\frac{A_{g,e,y}}{A_{g,e,2019}}\left(\frac{L_{g,e,y}}{L_{g,e,2019}}\right)^{1/\eta_e},$$

with $\eta_{\text{no BA}} = 4$ and $\eta_{\text{BA}} = 2$, and $L_{g,e,2019}$ at the fitted 2019 baseline.

5 Parents' employment paths

1. **Estimate relative local employment rates.** For each sex and fine age cell (16–17, 18–19, 20–24, and five-year bins from 25–29 through 55–59), I pool two adjacent ACS years and calculate the person-weighted employment rate. I divide the local rate by the national rate from the same pool. Pooling reduces noise in small local cells.

2. **Apply the national projection.** I multiply the corresponding national CPS/CBO projected employment rate by this local-to-national ratio. Observed ratios run through the 2023–2024 pool; I hold that last ratio fixed afterward. The path combines the national forecast with each place's latest relative position.

3. **Translate the path to caregiver targets.** I combine fine age cells into the model's under-25, 25–34, 35–44, and 45+ groups, separately by sex, using the geography's fixed 2019 ACS age-population weights. The 45+ path uses the available 45–59 projection cells. Each group's

target is its 2019 employment-forest rate times the growth in its projected local rate since 2019, capped below one. In each baseline year I fit the employment shift on working options to these targets and hold it fixed for that year's policy comparisons.

6 Implementation and checks

1. **Inputs built.** Demand and supply inputs cover all 52 geographies. Source and hash checks pass, and all 52 annual wage paths match their source data.

2. **Baseline fitting.** At local prices before the fit, the model returns P^0; after the fit, care hours match the targets to relative gaps below 10^{-8}. The New York target is the sum of the NYC target, on NYC's own income ranks, and a rest-of-state target, on New York's ranks. NY minus NYC therefore equals the rest of state exactly in every care type.

3. **Price validation.** I compare model full-time weekly prices with NDCP county median prices, including NDCP's filled-in values. County prices are weighted by Census population ages 0–4, and model prices by local predicted paid-center enrollment across tiers and ages. Model hourly prices become 40-hour weeks. Model ages 0–1, 2, and 3–4 match NDCP's infant (0–23 months), toddler (24–35), and preschool (36–54) groups.

   The local residual sets the 2019 model center price from NDCP, so the 2019 comparison is a check of the build, not a test. It covers the 46 states with an NDCP 2019 price in all three age groups: the unweighted correlation is 1.00, and the model price is 8.0% above NDCP in every state, the NSECE level factor.

   The 2020–2022 comparisons hold the 2019 local residuals fixed and grow them with CPI. They cover 46, 41, and 45 states. The unweighted correlations are 0.987, 0.981, and 0.977, and weekly RMSE is \$22.72, \$23.73, and \$33.03. The mean ratio of model to NDCP price is 1.105, 1.100, and 1.130.

   Paid homes use NDCP family child care prices, with ages weighted by model paid-home children. The 2019 comparison covers 45 states: the correlation is 1.00, and the model price is 36.6% above NDCP, the home level factor. For 2020–2022 it covers 45, 40, and 44 states, with correlations 0.985, 0.961, and 0.965 and mean ratios 1.381, 1.374, and 1.408.

   NDCP stops at 2022, so for 2026 I compare the full-sample run's equilibrium prices with 2022 NDCP prices grown with CPI. For centers, over 45 states, the correlation is 0.980 and the mean ratio is 1.154; for homes, over 44 states, 0.965 and 1.432. Both are above their level factors because the model's labor cost grows with wages, which rise faster than CPI. Equilibrium prices are within 2.5% of the 2019 cost formula at 2026 wages and CPI in every state.

4. **Simulation validation.** The check uses a 1% ACS sample for 2026–2027 across all 52 geographies: 104 geography-year solves. A solve must converge, have a paid-sector price gap below \$0.01 per child-hour, and miss each employment target by less than 0.001 percentage points. The 3 November run, with observed staffing, the summed New York target, and the 26 October calibration, passed all 104 solves. The largest paid-sector price gap was \$0.00032 per child-hour and the largest employment miss was 0.00030 percentage points. A full-sample run the same day, split into six jobs by geography, also passed all 104 solves, with a largest price gap of \$0.00017 and a largest employment miss of 0.00004 percentage points.

6.1 Open items

The five questions below come from the 3 November review.

1. **How do we adjust the local alphas? (3.4)** One utility shift per care sector and one per caregiver group, solved to match local care hours and employment. β and ρ stay national, because local data give care totals only. The drift check on the 3 November full-sample run measures the share of children ages 0–4 in paid centers, from the fitted 2019 baseline to the simulated 2026 baseline, over the 50 states and DC. The population-weighted mean change is -0.1 percentage points, and the mean absolute change is 0.5 points. State changes run from -2.0 to $+1.0$ points, or -1.9 to $+1.4$ with Gumbel-drawn choices. The 1% run's state moves of up to 18 points came from its small samples.

2. **How do NSECE, ACS, and BLS wages compare? (4.1)** My state wages are NSECE national wages times BLS OEWS state-to-national wage ratios, with the occupations mixed at national ACS shares. Nationally they match NSECE: \$11.86 against \$12.06 per hour without a BA, \$19.87 against \$19.70 with one. By Census region I compare three sources, each relative to its own national mean:

   - NSECE: center lead and assistant teachers' wages, with design standard errors.
   - Model (BLS OEWS): my state wages, averaged with ACS center care-worker hours by state of work as weights. Children ages 0–4 as weights give the same result within 2%.
   - ACS 2015–2019: hourly earnings of center care workers (employees in child day care services), in 2019 dollars, about 63,600 unweighted records.

Education	Region	NSECE (SE)	Model (OEWS)	ACS
No BA	Northeast	1.00 (0.04)	1.13	1.08
	Midwest	0.95 (0.03)	0.95	0.95
	South	0.93 (0.03)	0.91	0.94
	West	1.23 (0.05)	1.11	1.08
BA	Northeast	0.97 (0.07)	1.12	1.12
	Midwest	1.10 (0.11)	0.95	0.93
	South	0.96 (0.05)	0.90	0.94
	West	1.00 (0.06)	1.08	1.05

   Three cells differ from NSECE by more than two standard errors: the Northeast for both education groups and the West without a BA. In all three, the ACS agrees with the model, not with NSECE. The two large independent surveys agree, so the regional wage pattern stays on OEWS.

3. **How do we make the non-labor residual local? (4.3)** The residual is the local NDCP price minus the local labor cost, so 2019 local prices match NDCP.

4. **How do we make staffing reflect actual staffing, not legal requirements? (4.2)** Staffing comes from NSECE observed staffing by region and urban density, at each geography's density mix. With the local residual, staffing does not change 2019 prices; it sets the split between labor and non-labor cost, and so how prices respond to wages.

5. **Why don't we use CCDF more? (4.2)** In CCDF FY2019, 84% of CCDF-subsidized children in home-based care are in licensed homes, against 26% of all paid home-based children in NSECE. The state share has a correlation of 0.05 with the state's legal home capacity. CCDF covers subsidized children only, so it does not measure the licensed share of all paid home care. CCDF stays a diagnostic.

`SOURCES.csv` lists source links and dates.